

Discussion 6 Worksheet

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Learning goals:

- Compute projections onto a vector and onto a subspace.
- Use projection matrices $P = A(A^\top A)^{-1}A^\top$.
- Set up and solve a simple least-squares line of best fit via normal equations.

Formulas:

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}, \quad \text{Projection matrix } P = A(A^\top A)^{-1}A^\top \quad (\text{if } A \text{ has full rank}).$$

Problem 1: Find the projection of the $\vec{u}$ onto $\vec{v}$ when $\vec{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$

Problem 2: Find the projection of the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto the space spanned by the vectors $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

in the standard basis. (This is fairly computationally intensive so feel free to use a graphing calculator [such as [desmos.com/matrix](https://www.desmos.com/matrix)]; but be able to do each of these operations by hand on an exam).

Problem 3: If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

and the inverse of A exists, then using the inverse formula of a 2x2 matrix,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

show that $A^{-1}A = I$.

Problem 4: Boston rent can be modeled as a linear function of **bedrooms** (x_1), **bathrooms** (x_2), and **nearby rats** (x_3) using the equation:

$$y = m_1x_1 + m_2x_2 + m_3x_3 + b.$$

Use this data to fit the model:

Apt	y (price \$)	x_1 (beds)	x_2 (baths)	x_3 (rats)
1	1500	1	1	200
2	600	0	1	50
3	7000	3	2	5
4	3200	2	1	10

Then predict the price of an apartment with 2 beds, 1 bath, and 5 nearby rats. (Set up this problem and solve using [desmos.com/matrix](https://www.desmos.com/matrix))

What do each of the unknowns m_1, m_2, m_3, b represent?

Problem 5: Fit the following points to the parabola $y = ax^2 + bx + c$: $(-3,4)$, $(0,0)$, $(4,4)$.